

A review on machine learning techniques for text classification

M.Kavitha ¹, Research Scholar, Dr.P.Prabhavathy ², Assistant Professor

Department of Computer Science and Engineering, SRM Institute of Science and Technology, Vadapalani, India

¹ km2693@srmist.edu.in, ² prabhavp1@srmist.edu.in

Abstract

As the volume of information and documents available on the internet grows, so does the demand to access it, and the need for text classification becomes evident. Due to unstructured data, most businesses today are unable to realise their full potential. It is difficult to extract valuable insights from unstructured text data from various sources such as social media, emails, conversations, webpages, surveys, and it is also hard and time-consuming process. Companies can automate business processes and make data-driven business choices in a quick and cost-effective manner using machine learning text categorization algorithms. Scalability and real-time analysis are made easier by text tagging.

Keywords:

Machine learning, Text categorization, Stemming, Lemmatization, Feature engineering

1. Introduction

Most of the text and documents from internet are unstructured data. To use it as input for models, it must be converted to structured data. The data must first be pre-processed to exclude any undesirable characters, symbols, or words. The data is then subjected to a feature engineering process. The model is then trained and evaluated. In this work, we discussed types of machine learning [1], different types of text classification, and some text classification feature engineering techniques such as word embedding, length analysis, word frequency, topic labelling, named entity recognition, and sentiment analysis. Various issues in text classification and assessment metrics are also discussed in the article.

1.1 Recent Works

The paper [5] classified BBC news articles using supervised machine algorithms and used count vectors and TF-IDF for feature selection. Multinomial logistic regression shows better performance compared to other algorithms. J. F. DeRose et al., [7] compared Bidirectional Encoder Representations from Transformers (BERT) pre-trained model with fine-tuned models for natural language understanding. The paper [8] implemented text emotion classification by extracting the features from Bi-LSTM and CNN models. The paper [9] illustrated that transformer-based models and fusion based models outperform other traditional models by varying training set size for classifying the text from social media. The paper [11] predicted proactive

personality by taking short-answer questions text and social media post text using various machine learning algorithms. In [13] the authors implemented single-layer multi-size filters using convolutional neural network for text classification and compared its performance with baseline models varying its size. The paper [14] proposed NER task on Bangla Language using Word2Vec and BERT embeddings.

2. Machine learning

The model can be built either using rule based or machine based. Machine learning based model predicts the result accurately and quite easier for any modifications. Machine learning is a subfield of natural language processing and it assist in classifying and analysing the text documents. In general, it is classified as supervised, unsupervised, and semi-supervised learning [1,4,10].

2.1 Supervised learning

In supervised machine learning [2], models use labelled dataset for training. The goal of a supervised learning algorithm is to determine a mapping function that will map the dependent variable to the independent variable. It is used for variety of applications like speech recognition, image categorization, fraud identification, spam detection and object recognition. Regression and classification are two forms of supervised machine learning issues.

Regression is a numerical method to find the association between the dependent and independent variables. It is applied to predict continuous variables like weather forecasting, market trends, and so on. Linear Regression, Logistic Regression, Lasso Regression, and Polynomial Regression are some of the most popular regression techniques.

Classification is categorizing the related facts into groups based on certain criteria. It works on categorical variables. Some of the supervised machine and deep learning algorithms are Naïve Bayes, Decision tree, Random Forest, Support Vector Machine (SVM) [5,18], bagging models, boosting models, Convolutional Neural Network (CNN) [13], Long Short-Term Memory (LSTM), Gated Recurrent unit (GRU), bidirectional recurrent neural network [8,15].

2.2. Unsupervised learning

Machines are trained using unlabelled datasets in unsupervised learning. Machines sift through data to

uncover hidden patterns and insights. Clustering and association are the two kinds of unsupervised machine learning problems. K-means clustering, Anomaly detection, K-Nearest Neighbours (KNN), Apriori algorithm, Principle Component Analysis and Hierarchical clustering are some of the unsupervised learning algorithms.

2.3. Semi-supervised learning

Semi-supervised learning is an approach that learns from small set of labelled data and a huge amount of unlabelled data during training. It is the mixture of supervised and unsupervised machine learning. The underlying approach is similar; data are clustered using an unsupervised learning algorithm and then by using the existing labelled data to label the rest of the unlabelled data.

3. Types of text classification

Text classification and text extraction are the common task in natural language processing. Text tagging or text categorization are other terms for it. It categorizes text into organized group by analysing text documents and assign pre-defined tags based on their content. Text classification comes under supervised machine learning techniques.

Classification problems can be classified as single label, multi-class and multi label [16]. To distinguish between spam and non-spam emails is an example of single label classification with two target values. In multi-class problem each instance is associated with one label among many classes. For example, classifying fruit or animal from given set of categories is a multi-class problem. The given input will be assigned only one label in the case of single and multi-class classification whereas in multi label problem each input is assigned one or more labels among many categories. A problem to classify research articles into different domains is a multi-label classification. A research article can come under different categories.

4. Steps in Text classification Process

Text categorization involves various processes [6] like data preparation, feature engineering, training the model and improving the performance of text classifiers which is shown in Figure 1.

4.1 Data pre-processing

It is a process of transforming unrefined data into an understandable format. It is the fundamental step in text categorization. Most often, the real-world data is often incomplete and inconsistent. Data pre-processing [3,19] is a used to address such issues. Some of the data pre-processing steps are removing blank rows in the dataset, lower casing, tokenization, removal of stop words, removing non-alpha text, stemming and lemmatization.

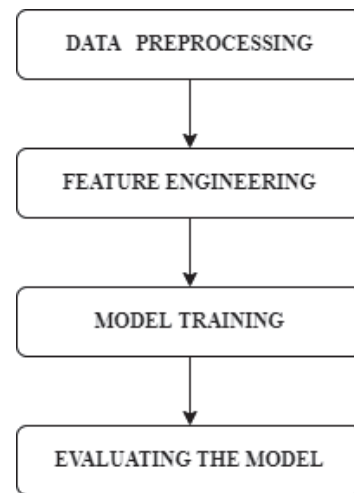


Fig. 1. Steps in text classification

Stop words are the most common words such as is, at, the, in, as, and on. Tokenization is splitting a raw text into words called tokens. It helps in further development of the model. It uses separation technique to split text into words called word tokenization. If we split text into sentences, it is called sentence tokenization.

Stemming [17] is the process used to convert the derived words into a root word. Stemming and lemmatization go hand in hand. The distinction is that a stemmer works on a single word without knowing the context, whereas lemmatization does so by knowing the context. As a result, stemmer is unable to differentiate between words that have various meanings depending on parts of speech they occur in. When compared to stemming, lemma performs better. Table I shows some instances for stemming and lemmatization process.

Table I
Stemming and Lemmatization

Form	Stem	Lemma
Babies	Babi	baby
Running	Runn	Run
Useful	Use	Useful
Beautifully	Beauti	Beautifully

4.2 Feature Engineering

Plain text cannot be processed by machine learning algorithms in their basic form. Natural language text is converted into numbers using word embedding techniques. It uses language modelling and feature extraction techniques to convert words to number vectors. Word2vec and GloVe are two most frequently used model for word embedding [12].

4.2.1 Vectorization Techniques

In feature engineering it is necessary to convert raw text into vector of features or words. Some of the general methods for vectorization are document vectorization or bag of words (BoW), one-hot encoding,

index-based encoding, counting vectors, TF-IDF vectors, N-Gram etc.

a. Document Vectorization

In document vectorization, we perform the steps like text pre-processing, creating vocabulary and creating a matrix of words with documents. Words are represented as columns and document as rows. A vocabulary is a list of all the unique words in the document. Feature vector is represented as $V[i, j] = 1$, if document i contains word j
 $V[i, j] = 0$, otherwise.

b. One-hot Encoding

The major drawback of document vectorization is it does not maintain occurrence of words. One-hot encoding retains the word order which is used in some machine learning algorithms like RNN and LSTM. An illustration for one-hot encoding [20] is given in Figure 2.

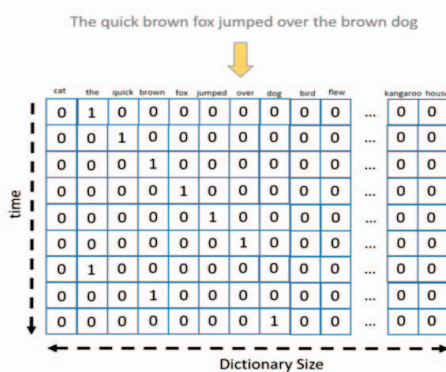


Fig. 2. One-hot Encoding

c. Index-based Encoding

Index-based encoding creates a dictionary that maps each word with one index. For example, consider the document "Feature extraction is transforming text data into vectors". The vocabulary for the above document is ['feature', 'extraction', 'is', 'transforming', 'text', 'data', 'into', 'vectors']. Index for the above document is [2,3,5,7,6,1,4,8]. The vector representation of "data" is [0.0,0.0,0.1,0,0]. This is a very basic approach for representing a word in vector form.

d. Counting Vector

The Corpus C containing D documents and N unique tokens extracted from it will have dictionary with N words, and $D \times N$ will determine the size of the Count Vector matrix M . The frequency of words in document $D(i)$ is represented in each row of the matrix M . CountVectorizer tool in python can be used to tokenize, build a vocabulary, and then encode a document. It calculates the frequency of words but does not find the important words in the document. Hence, TF-IDF is used to retrieve the important words in the document.

e. TF-IDF Encoding

TF-IDF is Term Frequency- Inverse Document Frequency [18]. TF is a technique to compute the weight for each word in the document and corpus. It is computed as shown in Equation 1.

$$TF(w) = \frac{\text{No. of times word } w \text{ occurs in a document}}{\text{Total number of words in a document}} \quad (1)$$

Inverse document frequency measures how important the term is. TF-IDF technique is frequently used in information retrieval, data mining, stop words filtering in text summarisation and classification. Let us consider a document with hundred words in that the term 'man' appears 6 times. Using Equation 1,

$$TF(\text{man}) = 6/100 = 0.06$$

$$IDF = \log(N/n) \quad (2)$$

In Equation 2, N correspond to the total number of documents and n denotes the total number of documents containing the word 'w'. If there are 10 million documents and the word 'man' appears 1000 times then,

$$IDF = \log\left(\frac{1,000,000}{1000}\right) = 3$$

thus TF-IDF is a product of these quantities, as shown in Equation 3.

$$TF-IDF(w) = TF(w) * IDF(w) \quad (3)$$

$$TF-IDF(\text{man}) = 0.06 * 3 = 0.18$$

f. N-Gram Language Modelling

Feature engineering [3] can be done by analysing and processing the whole text or single word. The N-Gram language modelling can be used to determine the significance of a particular word in a document. A sequence of n words from a text is called an n -gram. When the n -gram has a size of one, it is called a unigram, and when it has a size of two, it is called a bigram, and so on. For instance, the phrase "I like this movie" can be broken down into:

Unigrams: "I", "like", "this", "movie"

Bigrams: "I like", "like this", "this movie"

4.2.2 Feature extraction Techniques

Feature extraction is collecting required features from the dataset in order to reduce the effort to build the model. It also aims to reduce the number of features by using dimensionality reduction techniques [6]. We can analyse the length, sentiment and assign named entity to improve the performance of text classification.

a. Length Analysis

It is necessary to determine the length of the text because it can give a lot of insights for natural language processing. This feature helps to build the model easily. So, we need to count characters, words, sentences in the documents. Similarly finding average word length and average sentence length will be useful to achieve the task.

b. Sentiment Analysis

Finding sentiment or opinion mining [15,19] is one of the most difficult tasks due to the contextual words

and ambiguity in NLP. Because the context of the text is uncertain, this occurs. The best way would be to create our own sentiment model that is tailored to our specific data. When data is not available, pre-trained models like Textblob and Vader can be utilised instead. Textblob, which is based on NLTK and assigns polarity to words to assess the sentiment of the entire text as an average, is one of the most popular. Vader is Valence aware dictionary and sentiment reasoner which finds polarity as well as emotion intensity. It maps lexicon features with sentiment score that thrives with social media data.

c. Named-Entity Recognition (NER)

Named-entity recognition technique [14] is used for tagging named entities into pre-defined categories such as human names, organisations, locations, time expressions, and so on. Training a NER model takes a long time since it necessitates a large dataset. SpaCy is one of the most powerful open-source named-entity recognition tools available. It has a number of natural language processing (NLP) models that can recognise a wide range of objects.

4.2.3 Topic Analysis

Topic modelling and topic categorization are two types of topic analysis methodologies. A topic model used to identify hidden topics that arise in a set of documents. Topic modelling can be done with the Genism package. Topic modelling is a type of unsupervised machine learning that does not require any prior training. Topic classification is a supervised machine learning technique that requires training before it can be used to analyse texts automatically. Despite the additional groundwork required for topic categorization, this topic analysis strategy produces more accurate findings than unsupervised alternatives. We can gain more useful information that will help us make better decisions. When corporations want to know what customers think about specific elements of a product, they might use topic modelling. Rather than devoting a significant amount of time to analysis, we can infer topics of interest, with a topic modelling algorithm.

a. Latent Semantic Analysis (LSA)

LSA is a topic modelling technique which many data analysts make use of. As bag of words matrix or N-Gram matrix result in huge sparse matrix, we go for LSA technique which assist in reducing the matrix dimension. LSA use singular value decomposition to reduce the bag of word matrix. LSA calculates how frequently words appear in documents and over the entire corpus. The TF-IDF method is the industry standard for calculating word frequencies.

b. Latent Dirichlet Allocation (LDA)

LDA [3] is an unsupervised machine learning model used to extract topics. LDA and LSA both use bag of word matrix as input. Documents are usually represented as a jumble of latent topics, each of which is described by a word distribution. The hypothesis distribution (i.e., comparable themes utilise similar terms)

and the statistical distribution are the same fundamental assumptions in both LDA and LSA (i.e., documents that cover several topics). LDA is considered as two-layer aggregations. First layer is the distribution of categories and second layer is distribution of words within the category.

5. Challenges in classification

Some of the challenges [17] in text classification are analysing the semantics of text, diverse nature of text with highly variable quality and quantity, document with noise, ill-sampled data, and unknown classes. Problem may also occur due to large number of closely related classes. Sometimes stemming and lower casing may decline the performance.

5.1 Missing values

Data analyst face with many challenges, one among this is handling the missing values present in the dataset. To build a robust data model and make right decision we need to handle missing values. It can be handled by deleting rows that contain missing values, replacing the missing values, predicting the missing values to have a clear surge in performance.

5.2 Imbalanced dataset

The imbalanced dataset can degrade the performance of system. So, we need to balance the dataset before training. In general, there are two approaches under-sampling and oversampling. It is better to perform oversampling the data than under-sampling as it may lead to loss of information. Random sampling, synthetic minority oversampling technique (SMOTE), adaptive synthetic sampling (ADASYN) [21] are the approaches to balance the dataset.

5.3 Misclassification

Multi-word Expression, concordance, word sense disambiguation are the problems for misclassification of data. It can be solved by assigning PoS tag, finding word synonyms or semantic of the context and balancing the dataset.

6. Performance metrics

The purpose of text classification is to increase the performance metrics [9] by calculating accuracy, precision, recall and harmonic mean by overcoming the challenges in text tagging.

Accuracy

Accuracy is computed by finding the ratio of number of correct observations to total observations and it is given in Equation 4.

$$\text{Accuracy} = \frac{TP+TN}{(TP+TN+FP+FN)} \quad (4)$$

where TP, TN, FP and FN represent true positive, true negative, false positive, false negative values.

Confusion Matrix

Metrics are computed using confusion matrix. Table II shows the confusion matrix with event and no-

event. TP and TN are correctly predicted values. Let us consider a problem to understand aforementioned values. The problem is to predict whether a picture contains a Boy or Girl. Table III shows the test dataset containing 10 records with expected and predicted outcomes.

Table II
Confusion matrix with event and no-event

	Event	No-event
Event	TP	FP
No-event	FN	TN

Table III
Test dataset with expected and predicted values

	Expected	Predicted
1	Boy	Girl
2	Boy	Boy
3	Girl	Girl
4	Boy	Boy
5	Girl	Boy
6	Girl	Girl
7	Girl	Girl
8	Boy	Boy
9	Boy	Girl
10	Girl	Girl

The algorithm determines 7 correctly out of 10 with accuracy 70%.

Steps to create confusion matrix

1. Calculating the correct predictions for each class.
Boy classified as Boy - 3
Girl classified as Girl - 4
2. Calculating the incorrect predictions for each class.
Boy classified as Girl - 2
Girl classified as Boy - 1
3. Arranging these values into 2-class confusion matrix as shown in Table IV.

Table IV
Confusion matrix for the test dataset

	Boy	Girl
Boy	3	2
Girl	1	4

FScore/F1

It is calculated by finding precision and recall. Precision (quality) is the relation of true positives to all predicted positives given in Equation 6. Recall (quantity) is the number of true positive values divided by total number of actual positive values, provided in Equation 7. Precision(P) and recall(R) [9] have an inverse relationship. F1 or FScore is the harmonic mean used to

balance precision and recall and computed as shown in Equation 5.

$$FScore = \frac{2PR}{(P+R)} \quad (5)$$

$$P = \frac{TP}{(TP+FP)} \quad (6)$$

$$R = \frac{TP}{(TP+FN)} \quad (7)$$

Cross validation

Another way to increase performance is to use cross validation (CV) or rotation estimation. The learning dataset is separated into k distinct subsets of the same size in k-fold cross validation. The learning algorithm infers the model from each sample. The average performance of all the models is used to calculate overall performance. For computational reasons, smaller values of k (10 or 20) are desirable.

7. Applications of classification

Text classification is used in various applications [1] such as classifying tweets in social media, news headlines [5], chat bot queries, customer reviews, news articles, sentiment analysis, topic labelling, language detection, intent detection, document summarization, question answering, machine translation etc.

7.1 Machine Translation

The task of machine translation is to convert source language text into target language text. To perform this task the machine, need to understand and create semantic representation of the source language text as vector. Then vector is converted to target language. Machine translation assist user in several applications by translating technical manuals, product descriptions, emails, business documents and catalogues. The difficulty lies in determining the semantic meaning of sentences in order to deliver an accurate translation.

7.2 Automatic summarization

Summarization is a hot topic in artificial intelligence. The documents are summarised via automatic summarization. We can summarize either single document or multi document. It also aids in the comprehension of the emotional connotations included within the data, such as when gathering data from social media. Automatic summarising is particularly useful for providing a high-level summary of a news articles, user manuals, e-learning, legal documents, research, and medical cases

7.3 Question Answering (QA)

Search engines process natural language text and finds answers to the user queries which is still a challenging task. The demand for NLP arises as speech-to-text technology and voice-to-text applications develop. Question answering system [11] are used in chat boxes, education, customer support, virtual assistants and

data analytics. It can function as a text-only or spoken dialogue system. A question-answering application is a programme that can react rationally to a human request.

8. Conclusion

Text classification is the fundamental task in the field of artificial intelligence. It also acts as the foundation to data analytics, knowledge discovery, and information retrieval. In this paper, we highlighted the types of machine learning, steps involved in text classification, challenges and applications of text classification. From this paper we come to know performance of text classification depends on several factors such as corpus used, type of dataset, size of dataset, word embedding techniques, training time and models used for building.

References

- [1] Aljuaid, L., Wei, K. T., & Sharif, K. (2019). Machine learning: Tasks, modern day applications and challenges. *International Journal of Advanced Science and Technology*, 28(2), 329-340.
- [2] Kumar, R. R., Reddy, M. B., & Praveen, P. (2019). Text classification performance analysis on machine learning. *International Journal of Advanced Science and Technology*, 28(20), 691-697.
- [3] Conference on Information and Communication Technology (ICoICT). IEEE, 2018. Kowsari, Kamran, et al. "Text classification algorithms: A survey." *Information 10.4* (2019): 150
- [4] Thangaraj, M., & Sivakami, M. (2018). Text classification techniques: A literature review. *Interdisciplinary Journal of Information, Knowledge, and Management*, 117-135
- [5] Akanksha Patro, Mahima Patel, Richa Shukla and Dr. Jagurti Save, "Real Time News Classification Using Machine Learning", *IJAST*, vol. 29, no. 9s, pp. 620 - 630, May 2020.
- [6] M.M. Miro 'nczuk, J. Protasiewicz, 'A recent overview of the state-of-the-art elements of text classification', *Expert Systems With Applications* 106 (2018) 36–54, The Authors. Published by Elsevier Ltd
- [7] J. F. DeRose, J. Wang and M. Berger, "Attention Flows: Analyzing and Comparing Attention Mechanisms in Language Models," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 27, no. 2, pp. 1160-1170, Feb. 2021, doi: 10.1109/TVCG.2020.3028976.
- [8] M. Xuanyuan, L. Xiao and M. Duan, "Sentiment Classification Algorithm Based on Multi-Modal Social Media Text Information," in *IEEE Access*, vol. 9, pp. 33410-33418, 2021, doi: 10.1109/ACCESS.2021.3061450.
- [9] Al-Garadi, Mohammed & Yang, Yuan-Chi & Cai, Haitao & Ruan, Yucheng & Oconnor, Karen & Gonzalez, Graciela & Perrone, Jeanmarie & Sarker, Abeer. (2020). *Text Classification Models for the Automatic Detection of Nonmedical Prescription Medication Use from Social Media*. 10.1101/2020.04.13.20064089.
- [10] Aggarwal, C.C. *Machine Learning for Text*; Springer: Berlin/Heidelberg, Germany, 2018.
- [11] P. Wang et al., "Classification of Proactive Personality: Text Mining Based on Weibo Text and Short-Answer Questions Text," in *IEEE Access*, vol. 8, pp. 97370-97382, 2020, doi: 10.1109/ACCESS.2020.2995905.
- [12] Rezaeinia, S.M.; Ghodsi, A.; Rahmani, R. Improving the Accuracy of Pre-trained Word Embeddings for Sentiment Analysis. *arXiv* 2017, arXiv:1711.08609.
- [13] M. P. Akhter, Z. Jiangbin, I. R. Naqvi, M. Abdelmajeed, A. Mehmood and M. T. Sadiq, "Document-Level Text Classification Using Single-Layer Multisize Filters Convolutional Neural Network," in *IEEE Access*, vol. 8, pp. 42689-42707, 2020, doi: 10.1109/ACCESS.2020.2976744.
- [14] I. Ashrafi et al., "Banner: A Cost-Sensitive Contextualized Model for Bangla Named Entity Recognition," in *IEEE Access*, vol. 8, pp. 58206-58226, 2020, doi: 10.1109/ACCESS.2020.2982427.
- [15] Chen, T.; Xu, R.; He, Y.; Wang, X. Improving sentiment analysis via sentence type classification using BiLSTM-CRF and CNN. *Expert Syst. Appl.* 2017, 72, 221–230.
- [16] Sowmya B J, Chetan and K. G. Srinivasa, "Large scale multi-label text classification of a hierarchical dataset using Rocchio algorithm," 2016 International Conference on Computation System and Information Technology for Sustainable Solutions (CSITSS), 2016, pp. 291-296, doi: 10.1109/CSITSS.2016.7779373.
- [17] Singh, J.; Gupta, V. Text stemming: Approaches, applications, and challenges. *ACM Compu. Surv. (CSUR)* 2016, 49, 45.
- [18] Spirovski, K.; Stevanoska, E.; Kulakov, A.; Popeska, Z.; Velinov, G. Comparison of different model's performances in task of document classification. In *Proceedings of the 8th International Conference on Web Intelligence, Mining and Semantics*, Novi Sad, Serbia, 25–27 June 2018; p. 10.
- [19] Pahwa, B.; Taruna, S.; Kasliwal, N. Sentiment Analysis-Strategy for Text Pre-Processing. *Int. J. Comput. Appl.* 2018, 180, 15–18.
- [20] <https://www.kdnuggets.com/2019/11/text-encoding-review.html>
- [21] <https://www.analyticsvidhya.com/blog/2020/10/overcoming-class-imbalance-using-smote-techniques>